

Phase I/II Operating Characteristics

Method comparison - N = 60

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
BOIN12 - No. Pts treated	19.1	8.7	4	2.1	1	
BOIN12 - Select %	7.7	3.6	1.5	0.5	0.1	86.6
U-BOIN - No. Pts treated	17.3	10.9	9.1	7.6	6.7	
U-BOIN - Select %	12.3	18.3	16.6	16.7	12.7	23.4
EffTox - No. Pts treated	9.1	3.9	3.9	2.5	2.6	
EffTox - Select %	2	2	3	3	2	88
AIDE phase I (Utility 2) - No. Pts treated	20.3	8.3	2.3	0.5	0	
AIDE phase I (Utility 2) - Select %	4	5	2.1	0.1	0	88.8
AIDE phase I (Utility 3) - No. Pts treated	20.8	7.7	1.3	0.2	0	
AIDE phase I (Utility 3) - Select %	2.6	5.4	1.6	0.1	0	90.3
AIDE two-stage (Utility 2) - No. Pts treated	17.9	9.5	5.1	1.8	0.5	
AIDE two-stage (Utility 2) - Select %	14.4	5.4	1.5	0.1	0	78.6
AIDE two-stage (Utility 3) - No. Pts treated	17.9	9.5	5	1.8	0.5	
AIDE two-stage (Utility 3) - Select %	14.7	5.4	1.2	0.1	0	78.6
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
BOIN12 - No. Pts treated	17.2	15.9	18.6	6.1	1.6	
BOIN12 - Select %	25	22.2	43.2	7.6	0.4	1.6
U-BOIN - No. Pts treated	12.9	11.3	20	10.8	4.4	
U-BOIN - Select %	9.8	11	43.3	26.5	8.2	1.2
EffTox - No. Pts treated	43.3	3.8	3.6	3.6	4.5	
EffTox - Select %	70	5	7	9	7	2
AIDE phase I (Utility 2) - No. Pts treated	35.2	18.5	5.2	0.3	0	
AIDE phase I (Utility 2) - Select %	51.6	34	12	0.6	0	1.8
AIDE phase I (Utility 3) - No. Pts treated	39.8	16.4	2.8	0.1	0	
AIDE phase I (Utility 3) - Select %	60.7	30.3	6.7	0.2	0	2.1
AIDE two-stage (Utility 2) - No. Pts treated	7.8	12.3	21.1	12.9	5.2	
AIDE two-stage (Utility 2) - Select %	11.7	16	55.8	13.5	1.4	1.6
AIDE two-stage (Utility 3) - No. Pts treated	7.8	12.2	21.1	13	5.1	
AIDE two-stage (Utility 3) - Select %	16.5	16.4	51.9	12.4	1.2	1.6
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
BOIN12 - No. Pts treated	11.9	26.4	12	5.7	1.9	
BOIN12 - Select %	4.3	64.6	18.2	6	0.5	6.4
U-BOIN - No. Pts treated	9.2	25.7	13.4	7.4	3.7	
U-BOIN - Select %	2.8	53.7	22.4	13.2	6.9	1
EffTox - No. Pts treated	30.5	10.7	5.7	4.3	7.8	
EffTox - Select %	37	23	9	10	19	2
AIDE phase I (Utility 2) - No. Pts treated	15.6	39.5	3.1	0.1	0	
AIDE phase I (Utility 2) - Select %	7.8	82.5	5.7	0.2	0	3.8
AIDE phase I (Utility 3) - No. Pts treated	18	37.5	2.6	0.1	0	
AIDE phase I (Utility 3) - Select %	10.1	79.9	5.3	0.1	0	4.6
AIDE two-stage (Utility 2) - No. Pts treated	6.9	13.9	19.6	12.8	5.1	
AIDE two-stage (Utility 2) - Select %	3.3	47.4	33.7	9.8	1.4	4.4
AIDE two-stage (Utility 3) - No. Pts treated	6.9	13.9	19.6	12.8	5.1	
AIDE two-stage (Utility 3) - Select %	6.2	53.7	28.1	7.3	0.5	4.2

Phase I/II Operating Characteristics

Method comparison - N = 60

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
BOIN12 - No. Pts treated	9.6	11.3	14.7	18.2	5.8	
BOIN12 - Select %	6.2	12.4	28.1	44.8	6.7	1.8
U-BOIN - No. Pts treated	5.9	8.2	14.4	20.7	10.9	
U-BOIN - Select %	3.9	5.2	18.3	51.1	21.4	0.1
EffTox - No. Pts treated	11.2	7.8	12.7	10.3	16.9	
EffTox - Select %	2	7	26	31	32	2
AIDE phase I (Utility 2) - No. Pts treated	11.3	15.7	22.8	9.1	0.5	
AIDE phase I (Utility 2) - Select %	5.3	15.2	52.8	24.9	1	0.8
AIDE phase I (Utility 3) - No. Pts treated	14.5	20.2	18.8	5.8	0.3	
AIDE phase I (Utility 3) - Select %	5.4	23.7	49.2	17.9	0.6	3.2
AIDE two-stage (Utility 2) - No. Pts treated	3.7	4	10.9	23.6	17.8	
AIDE two-stage (Utility 2) - Select %	3	3.2	14	62.9	16.7	0.2
AIDE two-stage (Utility 3) - No. Pts treated	3.7	4	10.9	23.5	17.9	
AIDE two-stage (Utility 3) - Select %	4.5	5	14.8	61.4	14.1	0.2
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
BOIN12 - No. Pts treated	8.2	20	14.7	10.8	6	
BOIN12 - Select %	1.3	46.9	28.9	16.3	5.9	0.7
U-BOIN - No. Pts treated	6.3	18.9	15.9	11.6	7.4	
U-BOIN - Select %	0.9	41.8	28.9	18.5	9.9	0
EffTox - No. Pts treated	16.9	13.1	10.2	5.4	12.9	
EffTox - Select %	11	29	18	8	31	4
AIDE phase I (Utility 2) - No. Pts treated	8.3	40.4	9	1.9	0.3	
AIDE phase I (Utility 2) - Select %	0.3	76.3	18.2	4.3	0.7	0.2
AIDE phase I (Utility 3) - No. Pts treated	10.1	41	7.6	1	0.1	
AIDE phase I (Utility 3) - Select %	0.7	79.8	16.5	2.3	0.3	0.4
AIDE two-stage (Utility 2) - No. Pts treated	4	6.4	8.8	14.6	26.2	
AIDE two-stage (Utility 2) - Select %	3.5	33.2	23.7	20.5	18.9	0.2
AIDE two-stage (Utility 3) - No. Pts treated	4	6.5	8.8	14.2	26.4	
AIDE two-stage (Utility 3) - Select %	6.7	45	24.1	15.6	8.4	0.2
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
BOIN12 - No. Pts treated	34.3	10.6	4.6	2	0.7	
BOIN12 - Select %	61.9	12.8	4.3	0.8	0.2	20
U-BOIN - No. Pts treated	21.3	11.8	8.8	7.7	4.6	
U-BOIN - Select %	33.4	18.8	15.9	14.4	6.2	11.3
EffTox - No. Pts treated	51	1.3	0.8	0.7	0.7	
EffTox - Select %	84	1	2	1	0	12
AIDE phase I (Utility 2) - No. Pts treated	47.2	6	0.8	0.1	0	
AIDE phase I (Utility 2) - Select %	76.3	8.7	1.1	0	0	13.9
AIDE phase I (Utility 3) - No. Pts treated	48.3	5.5	0.2	0	0	
AIDE phase I (Utility 3) - Select %	78.6	7	0.3	0	0	14.1
AIDE two-stage (Utility 2) - No. Pts treated	29.5	14	7.3	2.5	0.6	
AIDE two-stage (Utility 2) - Select %	59.8	20.6	5.1	0.5	0	14
AIDE two-stage (Utility 3) - No. Pts treated	29.6	14	7.2	2.6	0.6	
AIDE two-stage (Utility 3) - Select %	63.3	19.1	3.6	0.2	0	13.8

Phase I/II Operating Characteristics

Method comparison - N = 60

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
BOIN12 - No. Pts treated	10.6	26.8	13.4	5.8	2.4	
BOIN12 - Select %	5.3	59.4	24.3	6.6	2.3	2.1
U-BOIN - No. Pts treated	9.8	25.8	14.1	6.4	2.9	
U-BOIN - Select %	2.9	54.3	26	9.9	5.1	1.8
EffTox - No. Pts treated	44.1	4.6	3.3	2.5	5.1	
EffTox - Select %	70	7	6	5	12	1
AIDE phase I (Utility 2) - No. Pts treated	29.6	28.1	1.4	0.1	0	
AIDE phase I (Utility 2) - Select %	38.6	56.6	2.7	0.1	0	2
AIDE phase I (Utility 3) - No. Pts treated	32.4	25.7	1	0	0	
AIDE phase I (Utility 3) - Select %	44.3	51.7	1.8	0	0	2.2
AIDE two-stage (Utility 2) - No. Pts treated	7.4	11.5	16.5	14.3	9.5	
AIDE two-stage (Utility 2) - Select %	5.1	41.8	30.7	15.4	5.2	1.8
AIDE two-stage (Utility 3) - No. Pts treated	7.5	11.6	16.5	14.1	9.5	
AIDE two-stage (Utility 3) - Select %	6.9	48.9	26.4	13.2	2.8	1.8
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
BOIN12 - No. Pts treated	25.9	8.7	3.8	2	0.7	
BOIN12 - Select %	23	8	2.2	0.2	0.2	66.4
U-BOIN - No. Pts treated	15.4	9.7	8.9	8.3	8.4	
U-BOIN - Select %	20	21.3	17.4	11.6	8.7	21
EffTox - No. Pts treated	19.1	5.4	4.6	3.1	3.9	
EffTox - Select %	14	8	11	5	3	58
AIDE phase I (Utility 2) - No. Pts treated	27.9	8.8	1.3	0.2	0	
AIDE phase I (Utility 2) - Select %	22	9.1	1	0	0	67.9
AIDE phase I (Utility 3) - No. Pts treated	28.8	7.9	0.7	0	0	
AIDE phase I (Utility 3) - Select %	21.2	8.4	0.7	0	0	69.7
AIDE two-stage (Utility 2) - No. Pts treated	23.1	10.3	4.5	1.3	0.3	
AIDE two-stage (Utility 2) - Select %	25.4	10.3	2.1	0.6	0	61.6
AIDE two-stage (Utility 3) - No. Pts treated	23.1	10.1	4.5	1.3	0.4	
AIDE two-stage (Utility 3) - Select %	26.4	9.8	1.5	0.4	0	61.9
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
BOIN12 - No. Pts treated	12.3	11.9	11.5	8.2	12.3	
BOIN12 - Select %	7.7	9.7	25.9	11.1	28.8	16.8
U-BOIN - No. Pts treated	8.6	7.8	11	10.1	21.6	
U-BOIN - Select %	2.4	3	13.7	15	64.3	1.6
EffTox - No. Pts treated	10.5	5.6	8.1	4.6	27.1	
EffTox - Select %	1	3	11	5	73	8
AIDE phase I (Utility 2) - No. Pts treated	14.8	15.3	18.6	5.9	1.3	
AIDE phase I (Utility 2) - Select %	5.2	15.1	49.4	15.1	3.6	11.6
AIDE phase I (Utility 3) - No. Pts treated	17.5	18.7	16.1	2.8	0.3	
AIDE phase I (Utility 3) - Select %	5.9	20.7	47.2	8.8	0.9	16.5
AIDE two-stage (Utility 2) - No. Pts treated	6.6	6	11.2	13.3	20	
AIDE two-stage (Utility 2) - Select %	4.5	5.9	19.8	17.7	43.3	8.8
AIDE two-stage (Utility 3) - No. Pts treated	6.6	6.1	11.2	13.2	20	
AIDE two-stage (Utility 3) - Select %	9.4	8.4	20.5	15.8	37.3	8.6

Phase I/II Operating Characteristics

Method comparison - N = 60

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
BOIN12 - No. Pts treated	13.5	21.6	9.3	4	1.6	
BOIN12 - Select %	3.6	45.1	14.4	2.7	0.7	33.5
U-BOIN - No. Pts treated	10	18	15	8	6.3	
U-BOIN - Select %	1.7	33.2	30.3	14.9	14.3	5.6
EffTox - No. Pts treated	19	10.9	6.4	5.9	12.4	
EffTox - Select %	10	21	13	16	27	12
AIDE phase I (Utility 2) - No. Pts treated	14.2	31.4	2.3	0.3	0	
AIDE phase I (Utility 2) - Select %	1.7	57.7	4	0.4	0	36.2
AIDE phase I (Utility 3) - No. Pts treated	15.1	30.9	1.4	0.1	0	
AIDE phase I (Utility 3) - Select %	2	58.6	2.8	0.2	0	36.4
AIDE two-stage (Utility 2) - No. Pts treated	10.8	15.4	13.5	6.4	2.5	
AIDE two-stage (Utility 2) - Select %	3.4	40.3	19.2	4.4	0.7	32
AIDE two-stage (Utility 3) - No. Pts treated	10.8	15.2	13.5	6.4	2.5	
AIDE two-stage (Utility 3) - Select %	4.2	41.9	17.3	3.7	0.4	32.5
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
BOIN12 - No. Pts treated	33.6	15.4	5	1.8	0.5	
BOIN12 - Select %	67.4	18.9	2.6	0.2	0	10.9
U-BOIN - No. Pts treated	28.9	13.5	7.2	4.4	3.8	
U-BOIN - Select %	64.2	22.4	6.3	1.2	1.1	4.8
EffTox - No. Pts treated	44.9	2.9	2.1	0.9	0.5	
EffTox - Select %	73	3	2	1	0	21
AIDE phase I (Utility 2) - No. Pts treated	44	11.2	1.1	0.1	0	
AIDE phase I (Utility 2) - Select %	70	18.4	1.2	0	0	10.4
AIDE phase I (Utility 3) - No. Pts treated	46.3	9.6	0.5	0	0	
AIDE phase I (Utility 3) - Select %	73.8	15.7	0.5	0	0	10
AIDE two-stage (Utility 2) - No. Pts treated	24.6	22.3	8.5	1.2	0.1	
AIDE two-stage (Utility 2) - Select %	59.5	29.1	2.6	0.1	0	8.7
AIDE two-stage (Utility 3) - No. Pts treated	24.5	22.2	8.6	1.2	0.1	
AIDE two-stage (Utility 3) - Select %	64.3	24.7	1.9	0	0	9.1
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
BOIN12 - No. Pts treated	11.1	13.5	11.8	15.9	7.4	
BOIN12 - Select %	11	20	13.3	45.8	7.4	2.5
U-BOIN - No. Pts treated	6.9	10.2	10.3	18.7	13.9	
U-BOIN - Select %	7.4	12.2	8.6	45.8	25.6	0.4
EffTox - No. Pts treated	10.5	6.3	14	11.9	15.5	
EffTox - Select %	1	4	26	39	26	5
AIDE phase I (Utility 2) - No. Pts treated	12.5	18.5	11.7	15.9	0.9	
AIDE phase I (Utility 2) - Select %	6.3	23.5	20.9	46.5	1.7	1.1
AIDE phase I (Utility 3) - No. Pts treated	16.1	22.3	10.8	9.5	0.4	
AIDE phase I (Utility 3) - Select %	8.6	28.5	24.3	33.6	1.1	3.9
AIDE two-stage (Utility 2) - No. Pts treated	3.9	4.9	9.8	25	16.4	
AIDE two-stage (Utility 2) - Select %	7.5	8.1	7	59.8	17.3	0.3
AIDE two-stage (Utility 3) - No. Pts treated	3.9	4.9	9.8	25	16.4	
AIDE two-stage (Utility 3) - Select %	12.1	11.5	5.5	55	15.5	0.4

Phase I/II Operating Characteristics

Method comparison - N = 60

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
BOIN12 - No. Pts treated	16.2	20	8.6	4.3	1.7	
BOIN12 - Select %	3.8	41.4	14.9	4	0.8	35.1
U-BOIN - No. Pts treated	10.6	16.2	14.8	9.2	7.5	
U-BOIN - Select %	2.8	32.2	34	15	12.4	3.6
EffTox - No. Pts treated	20.3	10.7	6	5.7	10.4	
EffTox - Select %	16	20	12	14	23	16
AIDE phase I (Utility 2) - No. Pts treated	14.3	31.2	3.9	0.4	0	
AIDE phase I (Utility 2) - Select %	2.7	58	7.5	0.6	0	31.2
AIDE phase I (Utility 3) - No. Pts treated	15.8	29.8	2.4	0.1	0	
AIDE phase I (Utility 3) - Select %	2	57.2	4.6	0.2	0	36
AIDE two-stage (Utility 2) - No. Pts treated	10.6	15.3	14.3	7.8	2.3	
AIDE two-stage (Utility 2) - Select %	5.5	39	20.2	7.3	0.2	27.8
AIDE two-stage (Utility 3) - No. Pts treated	10.6	15.2	14.3	7.8	2.3	
AIDE two-stage (Utility 3) - Select %	7.5	42.5	16.2	5.2	0.1	28.5
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
BOIN12 - No. Pts treated	14.3	14.9	12	11.1	6.2	
BOIN12 - Select %	13.9	19.3	16.5	30.3	9.6	10.4
U-BOIN - No. Pts treated	9.6	10.8	9.9	16	13.7	
U-BOIN - Select %	7.8	11.8	11.4	38.5	30.4	0.1
EffTox - No. Pts treated	10.2	5.7	13.1	8.6	15.8	
EffTox - Select %	1	3	24	21	35	16
AIDE phase I (Utility 2) - No. Pts treated	14.9	17.2	13.6	11.5	1.3	
AIDE phase I (Utility 2) - Select %	7.9	19.4	31.9	34.3	2.9	3.6
AIDE phase I (Utility 3) - No. Pts treated	17.5	20.3	11.8	7.7	0.6	
AIDE phase I (Utility 3) - Select %	7.5	22.5	32.1	27.3	1.4	9.2
AIDE two-stage (Utility 2) - No. Pts treated	5.2	7	12.1	18.8	16.2	
AIDE two-stage (Utility 2) - Select %	11.3	11	12.6	41.5	20.1	3.5
AIDE two-stage (Utility 3) - No. Pts treated	5.1	6.8	12.1	19	16.2	
AIDE two-stage (Utility 3) - Select %	19.6	15.5	10.9	36	14.3	3.7
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
BOIN12 - No. Pts treated	21.9	18.2	10.4	5	2.3	
BOIN12 - Select %	41.7	26.9	12.3	5.1	1.8	12.2
U-BOIN - No. Pts treated	21.5	16.5	9.8	6.8	5.2	
U-BOIN - Select %	41.3	29	13.7	9.8	5.4	0.8
EffTox - No. Pts treated	20.8	5.4	6.2	3.3	3	
EffTox - Select %	14	7	10	5	5	59
AIDE phase I (Utility 2) - No. Pts treated	26.2	20	8.8	2.2	0.4	
AIDE phase I (Utility 2) - Select %	28.9	36.1	21	3.2	0.3	10.5
AIDE phase I (Utility 3) - No. Pts treated	29.4	20	6.6	1.4	0.1	
AIDE phase I (Utility 3) - Select %	28.6	35.1	20.9	3.8	0	11.6
AIDE two-stage (Utility 2) - No. Pts treated	10.2	15.8	19.1	9.6	3.5	
AIDE two-stage (Utility 2) - Select %	40	30.5	15.7	4.1	0.1	9.6
AIDE two-stage (Utility 3) - No. Pts treated	10.1	16	18.9	9.8	3.4	
AIDE two-stage (Utility 3) - Select %	48.4	29.7	10.7	1.9	0	9.3

Phase I/II Operating Characteristics

Method comparison - N = 60

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
BOIN12 - No. Pts treated	15.1	14.5	12.7	10.9	6.8	
BOIN12 - Select %	34.2	26.9	20.5	13.5	4.9	0
U-BOIN - No. Pts treated	13	12	12.5	12.7	9.8	
U-BOIN - Select %	31.9	24	22.5	16	5.6	0
EffTox - No. Pts treated	41.6	3.8	5.3	3.4	4	
EffTox - Select %	67	5	10	7	6	5
AIDE phase I (Utility 2) - No. Pts treated	31.7	16.8	7.1	3.5	1	
AIDE phase I (Utility 2) - Select %	48.6	29.1	12.5	7.7	2.1	0
AIDE phase I (Utility 3) - No. Pts treated	34.4	17	6.1	2.1	0.4	
AIDE phase I (Utility 3) - Select %	52.6	30	12	4.7	0.7	0
AIDE two-stage (Utility 2) - No. Pts treated	6.1	5.6	6.4	9.4	32.5	
AIDE two-stage (Utility 2) - Select %	27.4	22.3	20	13.8	16.5	0
AIDE two-stage (Utility 3) - No. Pts treated	6.2	5.7	6.4	9.3	32.5	
AIDE two-stage (Utility 3) - Select %	37.1	26.3	18.3	11.7	6.6	0
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
BOIN12 - No. Pts treated	20.2	29.9	8.1	1.4	0.2	
BOIN12 - Select %	28.9	63.6	6.8	0.1	0	0.6
U-BOIN - No. Pts treated	18.7	28.4	10.1	2.1	0.6	
U-BOIN - Select %	25.7	58.6	12.9	1.3	1.1	0.4
EffTox - No. Pts treated	53.8	3.5	1.5	0.4	0.1	
EffTox - Select %	92	6	2	0	0	1
AIDE phase I (Utility 2) - No. Pts treated	40.8	18.4	0.7	0	0	
AIDE phase I (Utility 2) - Select %	62.8	35.8	1.1	0	0	0.3
AIDE phase I (Utility 3) - No. Pts treated	42.7	16.7	0.4	0	0	
AIDE phase I (Utility 3) - Select %	67.2	31.9	0.5	0	0	0.4
AIDE two-stage (Utility 2) - No. Pts treated	7.7	26.6	21.5	3.6	0.4	
AIDE two-stage (Utility 2) - Select %	10.5	68.9	19.8	0.4	0	0.4
AIDE two-stage (Utility 3) - No. Pts treated	7.8	26.7	21.3	3.6	0.3	
AIDE two-stage (Utility 3) - Select %	13.6	71.6	14.1	0.3	0	0.4
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
BOIN12 - No. Pts treated	9.8	12.6	12.6	11.5	13.6	
BOIN12 - Select %	10.5	20.4	22.4	16.4	30.2	0.1
U-BOIN - No. Pts treated	5.7	8.4	12.1	13.4	20.4	
U-BOIN - Select %	5.3	12	16.7	15.9	50.1	0
EffTox - No. Pts treated	10	5.4	14.5	9.4	18.5	
EffTox - Select %	1	3	29	20	41	6
AIDE phase I (Utility 2) - No. Pts treated	10.8	16.1	16.7	10.9	4.9	
AIDE phase I (Utility 2) - Select %	5.1	18.8	34.7	26.4	14.6	0.4
AIDE phase I (Utility 3) - No. Pts treated	13.3	19.5	15.8	8.1	2.5	
AIDE phase I (Utility 3) - Select %	7.1	24.5	38.1	22.1	7.1	1.1
AIDE two-stage (Utility 2) - No. Pts treated	3.4	3.3	4.9	12.3	36.1	
AIDE two-stage (Utility 2) - Select %	4.9	6.6	7.3	11.5	69.7	0
AIDE two-stage (Utility 3) - No. Pts treated	3.4	3.3	5	12.3	36	
AIDE two-stage (Utility 3) - Select %	9.6	9.5	9	10.4	61.5	0

Phase I/II Operating Characteristics

Method comparison - N = 60

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
BOIN12 - No. Pts treated	19	26.6	9.1	2.3	0.4	
BOIN12 - Select %	13.3	65.1	10.1	0.3	0	11.2
U-BOIN - No. Pts treated	13.9	24.9	12.1	5.2	3.4	
U-BOIN - Select %	14.5	59.9	18.2	2.9	2	2.5
EffTox - No. Pts treated	28.8	15.9	6.1	2.4	1.2	
EffTox - Select %	32	42	9	3	2	13
AIDE phase I (Utility 2) - No. Pts treated	15.4	38.2	4.6	0.3	0	
AIDE phase I (Utility 2) - Select %	5.6	81.8	8	0.2	0	4.4
AIDE phase I (Utility 3) - No. Pts treated	17.2	38.4	2.9	0.1	0	
AIDE phase I (Utility 3) - Select %	5.7	83.3	5.6	0	0	5.4
AIDE two-stage (Utility 2) - No. Pts treated	8.3	27.6	19.2	3.1	0.3	
AIDE two-stage (Utility 2) - Select %	7.2	69.1	18	0.2	0	5.5
AIDE two-stage (Utility 3) - No. Pts treated	8.3	27.5	19.3	3.1	0.3	
AIDE two-stage (Utility 3) - Select %	9.8	71.1	13.6	0.3	0	5.2
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
BOIN12 - No. Pts treated	13.4	26.2	12.9	5.3	1.9	
BOIN12 - Select %	4.8	73.3	16.9	2.8	0.4	1.8
U-BOIN - No. Pts treated	8.4	27.2	13.2	6.8	4.4	
U-BOIN - Select %	5.8	67.9	14.9	7.5	3.8	0.1
EffTox - No. Pts treated	19.4	17.6	11.3	3.4	2.8	
EffTox - Select %	14	43	22	4	3	14
AIDE phase I (Utility 2) - No. Pts treated	9.4	38.5	10.3	1.5	0.1	
AIDE phase I (Utility 2) - Select %	0.9	72.1	22.8	3.2	0.2	0.8
AIDE phase I (Utility 3) - No. Pts treated	11.1	40.7	7.3	0.6	0.1	
AIDE phase I (Utility 3) - Select %	1.3	81.1	15.5	1.2	0.1	0.8
AIDE two-stage (Utility 2) - No. Pts treated	4.4	12.6	26.3	12.9	3.6	
AIDE two-stage (Utility 2) - Select %	8.1	45.9	39.4	6.1	0.2	0.3
AIDE two-stage (Utility 3) - No. Pts treated	4.3	12.7	26.3	12.9	3.6	
AIDE two-stage (Utility 3) - Select %	11.4	53.2	30.9	4.1	0.1	0.3
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
BOIN12 - No. Pts treated	36.2	2.8	0.5	0	0	
BOIN12 - Select %	46.4	0.2	0	0	0	53.4
U-BOIN - No. Pts treated	33.9	5.4	2.3	1.6	1.4	
U-BOIN - Select %	47.9	4.6	2.5	1.5	1.7	41.8
EffTox - No. Pts treated	43	0.4	0.1	0	0	
EffTox - Select %	58	0	0	0	0	42
AIDE phase I (Utility 2) - No. Pts treated	39.7	0.5	0	0	0	
AIDE phase I (Utility 2) - Select %	46.1	0	0	0	0	53.9
AIDE phase I (Utility 3) - No. Pts treated	39.8	0.4	0	0	0	
AIDE phase I (Utility 3) - Select %	46.7	0	0	0	0	53.3
AIDE two-stage (Utility 2) - No. Pts treated	34.7	3.9	0.6	0.1	0	
AIDE two-stage (Utility 2) - Select %	42.3	0.5	0	0	0	57.2
AIDE two-stage (Utility 3) - No. Pts treated	34.6	3.9	0.7	0.1	0	
AIDE two-stage (Utility 3) - Select %	42.6	0.4	0	0	0	57

Phase I/II Operating Characteristics

Method comparison - N = 60

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
BOIN12 - No. Pts treated	21.3	27.1	6.5	2	0.3	
BOIN12 - Select %	19.6	64.6	3	0.4	0	12.4
U-BOIN - No. Pts treated	17.4	22.5	7.4	8.1	4.1	
U-BOIN - Select %	24.2	56	6.9	6.3	1.7	4.9
EffTox - No. Pts treated	27	15.7	5.8	1.8	0.7	
EffTox - Select %	26	41	6	1	1	26
AIDE phase I (Utility 2) - No. Pts treated	17.3	38.1	3.4	0.4	0	
AIDE phase I (Utility 2) - Select %	6.2	82	6.6	1	0	4.2
AIDE phase I (Utility 3) - No. Pts treated	19.2	37.3	2.5	0.1	0	
AIDE phase I (Utility 3) - Select %	7.1	84.8	4.4	0.1	0	3.6
AIDE two-stage (Utility 2) - No. Pts treated	9.2	34.8	13.2	1.8	0.2	
AIDE two-stage (Utility 2) - Select %	11.1	76.4	8.3	0.2	0	4
AIDE two-stage (Utility 3) - No. Pts treated	9.2	34.9	13.1	1.8	0.2	
AIDE two-stage (Utility 3) - Select %	15.9	74.7	5.5	0.2	0	3.7
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
BOIN12 - No. Pts treated	19.5	5.1	1.2	0.1	0	
BOIN12 - Select %	2.9	0	0	0	0	97.1
U-BOIN - No. Pts treated	19	8.4	3.9	2.9	2.7	
U-BOIN - Select %	4.3	2.3	1.6	0.3	0.1	91.4
EffTox - No. Pts treated	7.6	3.3	1.8	0.4	0	
EffTox - Select %	1	1	4	1	0	93
AIDE phase I (Utility 2) - No. Pts treated	19.5	4.1	0.3	0.1	0	
AIDE phase I (Utility 2) - Select %	1.3	1.7	0	0	0	97
AIDE phase I (Utility 3) - No. Pts treated	19.5	3.8	0.2	0	0	
AIDE phase I (Utility 3) - Select %	0.8	1.4	0	0	0	97.8
AIDE two-stage (Utility 2) - No. Pts treated	19.7	4.5	0.7	0	0	
AIDE two-stage (Utility 2) - Select %	2.8	0.4	0	0	0	96.8
AIDE two-stage (Utility 3) - No. Pts treated	19.8	4.5	0.7	0	0	
AIDE two-stage (Utility 3) - Select %	2.7	0.6	0	0	0	96.7
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
BOIN12 - No. Pts treated	4.4	4.9	7	30.7	13	
BOIN12 - Select %	0.2	0.6	4.8	73	21.4	0
U-BOIN - No. Pts treated	3.5	4.1	6.5	26.8	19.1	
U-BOIN - Select %	0.1	0.3	3.3	66	30.3	0
EffTox - No. Pts treated	9.1	4	7.4	9.3	29.9	
EffTox - Select %	0	1	8	20	70	1
AIDE phase I (Utility 2) - No. Pts treated	9.6	11.1	20.5	16.8	1.8	
AIDE phase I (Utility 2) - Select %	1.8	7.7	40.6	45	4.7	0.2
AIDE phase I (Utility 3) - No. Pts treated	11.6	12.7	19.6	14.5	1.2	
AIDE phase I (Utility 3) - Select %	3.9	8.7	42.4	41.6	2.9	0.5
AIDE two-stage (Utility 2) - No. Pts treated	3.1	3	3.5	9.1	41.3	
AIDE two-stage (Utility 2) - Select %	0.2	0.1	1.9	31.8	66	0
AIDE two-stage (Utility 3) - No. Pts treated	3.1	3	3.5	9.1	41.3	
AIDE two-stage (Utility 3) - Select %	1.7	0.8	4.2	38.2	55.1	0

Phase I/II Operating Characteristics

Method comparison - N = 60

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
BOIN12 - No. Pts treated	22.1	21.1	13.8	2.4	0.3	
BOIN12 - Select %	41.7	33.6	23.4	0.3	0	1
U-BOIN - No. Pts treated	18.5	19.1	15.3	4.3	2.7	
U-BOIN - Select %	37.9	31.5	24.3	3.3	2.9	0.1
EffTox - No. Pts treated	46.3	6.5	3.4	1.2	0.5	
EffTox - Select %	77	12	5	1	0	5
AIDE phase I (Utility 2) - No. Pts treated	33.5	20.8	5.4	0.2	0	
AIDE phase I (Utility 2) - Select %	48.5	38.1	13.1	0	0	0.3
AIDE phase I (Utility 3) - No. Pts treated	37.6	18.9	3.3	0.1	0	
AIDE phase I (Utility 3) - Select %	55.5	36.6	7.4	0.1	0	0.4
AIDE two-stage (Utility 2) - No. Pts treated	6.4	20.1	28.5	4.4	0.4	
AIDE two-stage (Utility 2) - Select %	19.8	32.5	46.6	0.8	0	0.3
AIDE two-stage (Utility 3) - No. Pts treated	6.3	20.2	28.5	4.4	0.4	
AIDE two-stage (Utility 3) - Select %	27.6	34.9	36.7	0.5	0	0.3
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
BOIN12 - No. Pts treated	19	19.6	14.9	3.4	0.4	
BOIN12 - Select %	16.1	31.2	40.5	1	0	11.2
U-BOIN - No. Pts treated	13.4	15.9	20.8	5.8	3.3	
U-BOIN - Select %	14.9	29.7	48.7	2.3	2.3	2.1
EffTox - No. Pts treated	27.1	12.8	9	3.8	1.9	
EffTox - Select %	27	32	20	5	2	14
AIDE phase I (Utility 2) - No. Pts treated	20.6	26.3	10.5	0.4	0	
AIDE phase I (Utility 2) - Select %	14	53.4	25.4	0.5	0	6.7
AIDE phase I (Utility 3) - No. Pts treated	23.6	28.2	5.5	0.3	0	
AIDE phase I (Utility 3) - Select %	16.3	61.5	14.8	0.1	0	7.3
AIDE two-stage (Utility 2) - No. Pts treated	8.4	16.8	26.7	5.6	0.4	
AIDE two-stage (Utility 2) - Select %	12.2	29.5	48.1	4.1	0	6.1
AIDE two-stage (Utility 3) - No. Pts treated	8.4	16.8	26.6	5.7	0.4	
AIDE two-stage (Utility 3) - Select %	16.1	33.4	40.8	3.8	0	5.9
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
BOIN12 - No. Pts treated	15.7	19.2	18.9	4.4	1	
BOIN12 - Select %	14.2	31.1	50	1.2	0	3.5
U-BOIN - No. Pts treated	10.8	15.9	21.4	7.8	4.1	
U-BOIN - Select %	11.7	27.2	48.9	8.1	3.6	0.5
EffTox - No. Pts treated	26.6	13.1	9	5.3	3.2	
EffTox - Select %	24	34	24	9	4	6
AIDE phase I (Utility 2) - No. Pts treated	17.9	27.3	13.7	0.5	0	
AIDE phase I (Utility 2) - Select %	11.8	50.6	35.3	0.6	0	1.7
AIDE phase I (Utility 3) - No. Pts treated	21.8	28.5	8.8	0.4	0	
AIDE phase I (Utility 3) - Select %	15.1	58.1	24.2	0.8	0	1.8
AIDE two-stage (Utility 2) - No. Pts treated	5.8	12.3	30.6	9.3	1.3	
AIDE two-stage (Utility 2) - Select %	7.6	18.6	67.1	4.7	0	2
AIDE two-stage (Utility 3) - No. Pts treated	5.9	12.3	30.6	9.3	1.3	
AIDE two-stage (Utility 3) - Select %	12.2	21.5	60	4.5	0	1.8

Phase I/II Operating Characteristics

Method comparison - N = 60

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
BOIN12 - No. Pts treated	18	23.8	5.3	1.3	0.3	
BOIN12 - Select %	5	52.9	2.6	0.1	0	39.4
U-BOIN - No. Pts treated	13.4	25.2	8.7	5.7	3.8	
U-BOIN - Select %	6.7	68.4	7.5	3.7	0.9	12.8
EffTox - No. Pts treated	23.5	15.9	5	2.4	1.5	
EffTox - Select %	21	42	5	2	1	30
AIDE phase I (Utility 2) - No. Pts treated	15.6	31.7	1.7	0.1	0	
AIDE phase I (Utility 2) - Select %	3	61.1	1.3	0.1	0	34.5
AIDE phase I (Utility 3) - No. Pts treated	16.3	31	1.1	0	0	
AIDE phase I (Utility 3) - Select %	2.1	60.2	1.6	0	0	36.1
AIDE two-stage (Utility 2) - No. Pts treated	13.3	26.6	8.4	1	0.1	
AIDE two-stage (Utility 2) - Select %	7.9	53.7	3.4	0	0	35
AIDE two-stage (Utility 3) - No. Pts treated	13.3	26.6	8.4	1	0.1	
AIDE two-stage (Utility 3) - Select %	9.2	53.2	3.1	0	0	34.5
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
BOIN12 - No. Pts treated	37	16.7	3.6	0.7	0.1	
BOIN12 - Select %	72.8	22.5	0.8	0	0	3.9
U-BOIN - No. Pts treated	35.6	16.8	4.1	1.3	1	
U-BOIN - Select %	69.4	26.2	0.9	0.8	0.4	2.3
EffTox - No. Pts treated	57.1	0.9	0.3	0.1	0	
EffTox - Select %	96	1	0	0	0	3
AIDE phase I (Utility 2) - No. Pts treated	52	6.9	0.1	0	0	
AIDE phase I (Utility 2) - Select %	86.5	11.6	0.1	0	0	1.8
AIDE phase I (Utility 3) - No. Pts treated	52.8	6.1	0.1	0	0	
AIDE phase I (Utility 3) - Select %	87.8	9.9	0.2	0	0	2.1
AIDE two-stage (Utility 2) - No. Pts treated	22.1	29.5	6.9	0.6	0	
AIDE two-stage (Utility 2) - Select %	53.7	41.9	2.5	0	0	1.9
AIDE two-stage (Utility 3) - No. Pts treated	22.1	29.4	6.9	0.6	0	
AIDE two-stage (Utility 3) - Select %	58.5	38.2	1.4	0	0	1.9
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
BOIN12 - No. Pts treated	12.5	14.3	12.7	16.1	4.5	
BOIN12 - Select %	16.7	20.6	15	45.4	1.9	0.4
U-BOIN - No. Pts treated	8.8	11.5	12.7	20.5	6.5	
U-BOIN - Select %	13.3	15.5	16.2	49.6	5.3	0.1
EffTox - No. Pts treated	25.8	8.1	11.5	8.2	5.5	
EffTox - Select %	26	16	28	22	6	2
AIDE phase I (Utility 2) - No. Pts treated	20.4	19.5	13.5	6.2	0.3	
AIDE phase I (Utility 2) - Select %	22.1	30.9	28.3	17.6	0.8	0.3
AIDE phase I (Utility 3) - No. Pts treated	24.9	21.4	9.2	4.2	0.2	
AIDE phase I (Utility 3) - Select %	27.6	38.2	21.6	12.3	0.2	0.1
AIDE two-stage (Utility 2) - No. Pts treated	3.8	4.1	10.2	29.7	12.1	
AIDE two-stage (Utility 2) - Select %	10.3	9.8	10	61.9	8	0
AIDE two-stage (Utility 3) - No. Pts treated	3.8	4.1	10.3	29.8	12.1	
AIDE two-stage (Utility 3) - Select %	17.6	14.7	10	51.1	6.6	0

Phase I/II Operating Characteristics

Method comparison - N = 60

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
BOIN12 - No. Pts treated	4.4	24.5	17.5	11.4	2.1	
BOIN12 - Select %	0.8	47.2	33.4	18.3	0.2	0.1
U-BOIN - No. Pts treated	5.2	18	17	15.8	4	
U-BOIN - Select %	1	42.4	34.2	22.3	0.1	0
EffTox - No. Pts treated	38.8	6	7.1	6.5	1.5	
EffTox - Select %	58	12	16	13	2	0
AIDE phase I (Utility 2) - No. Pts treated	26.1	31.2	2.6	0.2	0	
AIDE phase I (Utility 2) - Select %	32.2	62.6	4.9	0.3	0	0
AIDE phase I (Utility 3) - No. Pts treated	28.8	29.3	1.7	0.1	0	
AIDE phase I (Utility 3) - Select %	38.9	57.5	3.3	0.3	0	0
AIDE two-stage (Utility 2) - No. Pts treated	3.6	4.5	7	29.6	15.3	
AIDE two-stage (Utility 2) - Select %	3.1	33.2	25.1	33.2	5.4	0
AIDE two-stage (Utility 3) - No. Pts treated	3.6	4.6	7	29.5	15.3	
AIDE two-stage (Utility 3) - Select %	5.7	40	25	26.3	3	0
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
BOIN12 - No. Pts treated	10.1	13.8	20.4	11	4.8	
BOIN12 - Select %	5.9	16.4	57	15	5.4	0.3
U-BOIN - No. Pts treated	5.8	9.9	24.9	13	6.4	
U-BOIN - Select %	4.2	13.4	57.5	16.1	8.8	0
EffTox - No. Pts treated	16.8	10.8	16.9	6.6	7.3	
EffTox - Select %	7	20	46	11	12	4
AIDE phase I (Utility 2) - No. Pts treated	12.7	19.4	21.6	5.4	0.8	
AIDE phase I (Utility 2) - Select %	6.3	29	50.3	12.5	1.8	0.1
AIDE phase I (Utility 3) - No. Pts treated	15.1	22.6	18.4	3.5	0.3	
AIDE phase I (Utility 3) - Select %	7.9	38	44.1	9.1	0.5	0.4
AIDE two-stage (Utility 2) - No. Pts treated	3.5	3.9	9.8	19.8	23	
AIDE two-stage (Utility 2) - Select %	9.3	11.8	33.8	30.8	14.3	0
AIDE two-stage (Utility 3) - No. Pts treated	3.5	3.9	9.7	19.7	23.1	
AIDE two-stage (Utility 3) - Select %	16.3	19.3	34.7	23.6	6.1	0
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
BOIN12 - No. Pts treated	10.1	12.5	16	14.6	6.7	
BOIN12 - Select %	4.9	13	37.9	34.8	8.8	0.6
U-BOIN - No. Pts treated	4.8	6.7	17.8	17.7	12.9	
U-BOIN - Select %	3.1	5.9	29	30.9	31.1	0
EffTox - No. Pts treated	9.4	5.9	19	11.2	13.9	
EffTox - Select %	1	3	45	27	22	2
AIDE phase I (Utility 2) - No. Pts treated	10.2	12.1	21.5	14.3	1.5	
AIDE phase I (Utility 2) - Select %	3.8	9.1	43.6	39.3	3.9	0.3
AIDE phase I (Utility 3) - No. Pts treated	12.2	13.8	20.6	12.2	0.6	
AIDE phase I (Utility 3) - Select %	5.3	10.6	47.3	34.5	1.8	0.5
AIDE two-stage (Utility 2) - No. Pts treated	3.2	3.3	7.4	21.7	24.4	
AIDE two-stage (Utility 2) - Select %	5.8	3.9	11.1	49.7	29.5	0
AIDE two-stage (Utility 3) - No. Pts treated	3.2	3.2	7.5	21.8	24.4	
AIDE two-stage (Utility 3) - Select %	9.2	6.2	11.5	48.8	24.3	0

Phase I/II Operating Characteristics

Method comparison - N = 60

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
BOIN12 - No. Pts treated	15	17.3	16.3	7.2	3	
BOIN12 - Select %	13.1	22.3	40.8	12.3	2.9	8.6
U-BOIN - No. Pts treated	8.7	12.7	17.6	11.9	9	
U-BOIN - Select %	10.3	14.1	34.9	22.8	17.6	0.3
EffTox - No. Pts treated	13.7	12	15.4	5.8	8.8	
EffTox - Select %	4	20	35	12	17	12
AIDE phase I (Utility 2) - No. Pts treated	12.4	17.4	26	3.3	0.4	
AIDE phase I (Utility 2) - Select %	5.1	17.6	66.4	8.5	0.7	1.7
AIDE phase I (Utility 3) - No. Pts treated	15.1	20.7	21.2	2.2	0.1	
AIDE phase I (Utility 3) - Select %	5.5	24.3	58.6	7	0.3	4.3
AIDE two-stage (Utility 2) - No. Pts treated	4.6	8.3	23.2	15.3	8.5	
AIDE two-stage (Utility 2) - Select %	8.4	9.9	58.2	18	4.4	1.1
AIDE two-stage (Utility 3) - No. Pts treated	4.5	8.3	23.2	15.3	8.5	
AIDE two-stage (Utility 3) - Select %	13.3	15	53.5	15	2.1	1.1
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
BOIN12 - No. Pts treated	8.8	32.4	12.4	4.9	1.4	
BOIN12 - Select %	5.3	77.6	14.6	2.3	0.2	0
U-BOIN - No. Pts treated	7.4	31.6	14.6	5	1.3	
U-BOIN - Select %	6.2	70.8	17.4	4.1	1.5	0
EffTox - No. Pts treated	42.5	9.7	5	1.6	1	
EffTox - Select %	67	22	8	2	1	0
AIDE phase I (Utility 2) - No. Pts treated	27.8	29.3	2.8	0.1	0	
AIDE phase I (Utility 2) - Select %	37.2	56.9	5.8	0.1	0	0
AIDE phase I (Utility 3) - No. Pts treated	29.2	28.2	2.4	0.2	0	
AIDE phase I (Utility 3) - Select %	39.2	55.4	4.9	0.5	0	0
AIDE two-stage (Utility 2) - No. Pts treated	3.7	9.5	25.4	14.5	6.8	
AIDE two-stage (Utility 2) - Select %	6.8	41.1	41.2	9.7	1.2	0
AIDE two-stage (Utility 3) - No. Pts treated	3.7	9.6	25.4	14.4	6.9	
AIDE two-stage (Utility 3) - Select %	10.1	47.5	34.3	7.7	0.4	0
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
BOIN12 - No. Pts treated	17.7	21.7	8.1	4.4	1.8	
BOIN12 - Select %	8.6	47.5	11.5	5.7	1.2	25.5
U-BOIN - No. Pts treated	12.9	14.9	10.8	10.8	10.4	
U-BOIN - Select %	10.5	28.2	17.9	22.9	19.6	0.9
EffTox - No. Pts treated	17.8	11.4	8.6	4.2	8.4	
EffTox - Select %	11	22	16	7	16	28
AIDE phase I (Utility 2) - No. Pts treated	14.8	32.3	6.5	1	0.1	
AIDE phase I (Utility 2) - Select %	4	63.1	12	1.9	0.1	18.9
AIDE phase I (Utility 3) - No. Pts treated	16.4	32.4	5	0.6	0.1	
AIDE phase I (Utility 3) - Select %	3.6	65.5	11.2	0.9	0.1	18.7
AIDE two-stage (Utility 2) - No. Pts treated	9.7	21.3	15.2	6.5	3	
AIDE two-stage (Utility 2) - Select %	11.6	48.9	17.9	5.2	0.8	15.6
AIDE two-stage (Utility 3) - No. Pts treated	9.7	21.3	15.1	6.6	3	
AIDE two-stage (Utility 3) - Select %	14.5	51.7	13.9	3.9	0.4	15.6
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
BOIN12 - No. Pts treated	19.5	23.9	7.1	3	1.2	
BOIN12 - Select %	12.6	53.3	7.5	2.2	0.3	24.1
U-BOIN - No. Pts treated	16.3	19.2	10.2	7.5	6.4	
U-BOIN - Select %	15.8	43.1	17.1	12	8.2	3.8
EffTox - No. Pts treated	18.1	12.8	8.8	2.6	2.6	
EffTox - Select %	9	27	16	4	3	42
AIDE phase I (Utility 2) - No. Pts treated	14.6	36	6	0.9	0.1	
AIDE phase I (Utility 2) - Select %	4.1	72.3	10.6	1.1	0.1	11.8
AIDE phase I (Utility 3) - No. Pts treated	16.3	35.9	4.4	0.4	0	
AIDE phase I (Utility 3) - Select %	5	72	8.6	0.4	0	14
AIDE two-stage (Utility 2) - No. Pts treated	9.7	26.9	15	4.7	1.4	
AIDE two-stage (Utility 2) - Select %	15.5	60.6	12.7	0.8	0.1	10.3
AIDE two-stage (Utility 3) - No. Pts treated	9.6	26.8	15.1	4.7	1.4	
AIDE two-stage (Utility 3) - Select %	18.7	60.2	10.1	0.4	0	10.6